The Semiprime Neighbour on the $6N$ Skeleton: A Slower ω-Enrichment and Its Mechanism

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Abstract

Part XII established that twin centres enrich with the factor count $\omega_{>3}(N)$ through the single-prime law $P(q \mid N \mid \text{twin}) = 1/(q-2)$. Here we relax the right member from prime to *prime-or-semiprime*: we count centres with $6N-1$ prime and $6N+1$ either prime or a product of exactly two primes. This neighbour count also enriches with $\omega_{>3}(N)$, but systematically *more slowly* than the twin: on S_9 (3.5×10^6 centres with $6N-1$ prime at $\omega=1$), the twin rate rises to $1.81\times$ at $\omega=5$ while the exactly-semiprime part rises only to $1.37\times$. We give the mechanism: the small-prime avoidance that a factor-rich N imposes on $6N+1$ helps a prime (all-or-nothing) more than a semiprime (which may carry one small factor), and we confirm it directly—among semiprime $6N+1$, the fraction with a prime factor ≤ 97 falls from 0.56 at $\omega=1$ to 0.41 at $\omega=5$, so factor-rich N pushes $6N+1$ toward large$\times$large semiprimes. The local densities are the classical Hardy–Littlewood/Landau type; the contribution is the measured slower enrichment and its factor-structure mechanism. An exact double-factor CRT closed form is left open, as the semiprime's two-factor structure does not reduce to the clean single-prime $1/(q-2)$ of the twin. No claim is made about the infinitude of any pattern.

1 The relaxed neighbour

The twin counts centres N with both wings $6N\pm1$ prime; Part XII showed the count enriches with $\omega_{>3}(N)$ via $P(q \mid N \mid \text{twin}) = 1/(q-2)$. We relax the right wing: keep $6N-1$ prime, but allow $6N+1$ to be *prime or semiprime*, where a semiprime is a product of exactly two primes (with multiplicity, e.g. $25=5^2$, $35=5\cdot7$), i.e. $\Omega(6N+1)\in\{1,2\}$ with Ω the prime-factor count with multiplicity. This is the natural "Chen-type" relaxation of the twin condition, and we ask how its conditional density depends on $\omega_{>3}(N)$.

2 A slower enrichment

Binned by $\omega_{>3}(N)$ and normalised to $\omega=1$ (Table 1, Fig. 1 left), all three rates rise with ω, but at different speeds. Decomposing the relaxed rate into its prime part (the twin) and its exactly-semiprime part ($\Omega(6N+1)=2$):

Observation 1. *The semiprime neighbour enriches with $\omega_{>3}(N)$ monotonically but distinctly more slowly than the twin: at $\omega=5$ the twin reaches $1.81\times$ its $\omega=1$ rate, the exactly-semiprime part only $1.37\times$. The enrichment ratio (semiprime-only over twin) falls monotonically, from 1.00 at $\omega=1$ to 0.76 at $\omega=5$, so the semiprime does not simply scale the twin enrichment—it follows a genuinely flatter ω-law.*

ω	#($6N-1$ pr)	twin / ω=1	semi-nbr / ω=1	semiprime-only / ω=1
1	3,549,831	1.000	1.000	1.000
2	8,643,794	1.125	1.082	1.066
3	7,372,158	1.297	1.191	1.152
4	2,613,621	1.522	1.327	1.254
5	351,717	1.814	1.492	1.372

Table 1: Enrichment with $\omega_{>3}(N)$ on S_9, normalised to $\omega = 1$. The twin (right wing prime) rises fastest; the exactly-semiprime part (right wing semiprime) rises slowest; the relaxed prime-or-semiprime neighbour lies between.

3 The mechanism

The twin enrichment of Part XII is a small-prime avoidance effect: a factor-rich N (large ω) forces, through the dead residues $\mathrm{dead}(q) = \{\pm 6^{-1} \bmod q\}$, that $6N \pm 1$ avoid divisibility by the small primes $q \mid N$, raising the chance both wings are prime. For the right wing to be *prime* this avoidance is all-or-nothing and maximally helpful. For the right wing to be *semiprime*, it is less helpful: a semiprime is allowed to carry one small prime factor, so suppressing small factors removes some of the very semiprimes being counted. The enrichment is therefore weaker.

This predicts a direct signature: as ω grows, the semiprimes $6N + 1$ that survive should increasingly be products of two *large* primes rather than small×large. We confirm it (Fig. 1 right): among semiprime $6N + 1$ with $6N - 1$ prime, the fraction having a prime factor ≤ 97 falls from 0.56 at $\omega = 1$ to 0.41 at $\omega = 5$ (on S_9). Factor-rich N pushes its semiprime neighbour toward the large×large regime, exactly as the mechanism requires.

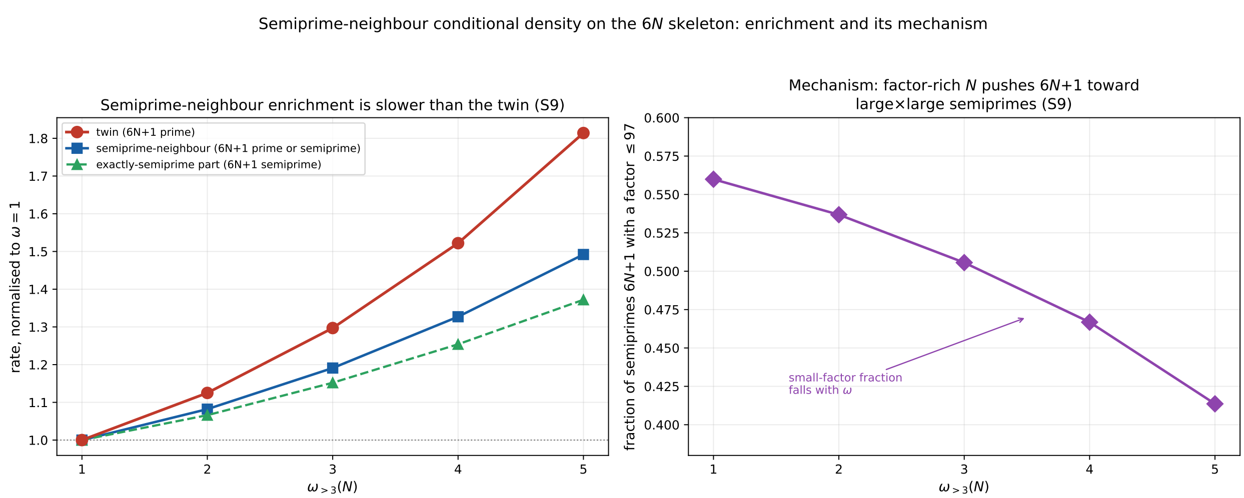

Figure 1: Left (S_9): enrichment of the twin (fastest), the semiprime-neighbour (intermediate), and the exactly-semiprime part (slowest). Right (S_9): among semiprime $6N + 1$, the fraction with a small prime factor (≤ 97) falls with ω (from 0.56 to 0.41), confirming the shift toward large×large semiprimes.

4 Scope and the open closed form

The single-prime and semiprime local densities are classical (Hardy–Littlewood for the prime wing; Landau's semiprime density for the relaxed wing); this paper introduces no new density formula. The contribution is the measured fact that the semiprime neighbour enriches more slowly than the twin, and the factor-structure mechanism for it, verified by the falling small-factor fraction.

We do *not* claim an exact closed form for the semiprime enrichment. The twin enrichment reduces to the clean single-prime law $1/(q-2)$ (Part XII), but the semiprime condition $\Omega(6N+1) = 2$ couples *two* prime factors, and the ω-dependence does not separate into a single per-prime factor; a faithful model would be a double-factor convolution that we find does not reduce to a clean product at the percent level. We therefore report the enrichment and its mechanism empirically and leave the exact double-factor closed form as an open problem, rather than force an approximate product. As throughout this series, no statement is made about the infinitude of twins, semiprime neighbours, or any pattern.

5 Conclusion

Relaxing the twin's right wing from prime to prime-or-semiprime yields a count that still enriches with $\omega_{>3}(N)$ but more slowly—to $1.37\times$ at $\omega = 5$ for the exactly-semiprime part, against $1.81\times$ for the twin. The mechanism is that a factor-rich N's small-prime avoidance helps a prime more than a semiprime, pushing surviving semiprimes toward large×large products, which we confirm directly. The exact double-factor closed form is left open. This places the semiprime neighbour alongside the cousin and sexy pairs as a relaxation of the twin whose conditional shape is set, qualitatively and mechanistically, by the same small-prime factor structure.

Data and code availability. The enrichment counts and the small-factor diagnostic are openly available [4].

References

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[4] R. Chen, $6N$ *semiprime neighbour: data and code*, `https://github.com/Ruqing1963/6N-semiprime-neighbour` (2026).